Lagrange Multipliers

Optimizing while an equality must hold

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A fixed budget

Two quantities, one budget

Choose two non-negative quantities x and y. Their product is the score, but the total is fixed:

$$\max_{x,y} \quad xy \quad \text{subject to} \quad x + y = 1, \quad x \geq 0, y \geq 0.$$

Some feasible choices:

x	y	xy
0	1	0
0.2	0.8	0.16
0.5	0.5	$\mathbf{0.25}$
0.8	0.2	0.16
1	0	0

The equal split looks best. We need a method that still works when substitution is not so easy.

Where this form appears in ML

“Maximize engagement subject to fairness” is the same shape: prefer one quantity while another is pinned.

choice	objective	equality
probabilities p_i	entropy or likelihood	$\sum_i p_i = 1$
recommender $\boldsymbol{\theta}$	engagement	fairness statistic $= 0$
portfolio weights w_i	predicted return	$\sum_i w_i = 1$
weights $\boldsymbol{w}$ after a step	$\|\boldsymbol{w} - \boldsymbol{z}\|^2$ (stay close)	$\|\boldsymbol{w}\|^2 = r^2$ (norm ball)
model parameters $\boldsymbol{\theta}$	training loss	$A\boldsymbol{\theta} = \boldsymbol{b}$
resource allocation x_i	utility	$\sum_i x_i = B$

The objective says what we prefer. The constraint says what is allowed.

Unconstrained and constrained stationarity

Without a constraint, a differentiable interior optimum satisfies

$$\nabla f(\boldsymbol{x}^*) = \mathbf{0}.$$

With $h(\boldsymbol{x}) = 0$, the optimum may have $\nabla f(\boldsymbol{x}^*) \neq \mathbf{0}$.

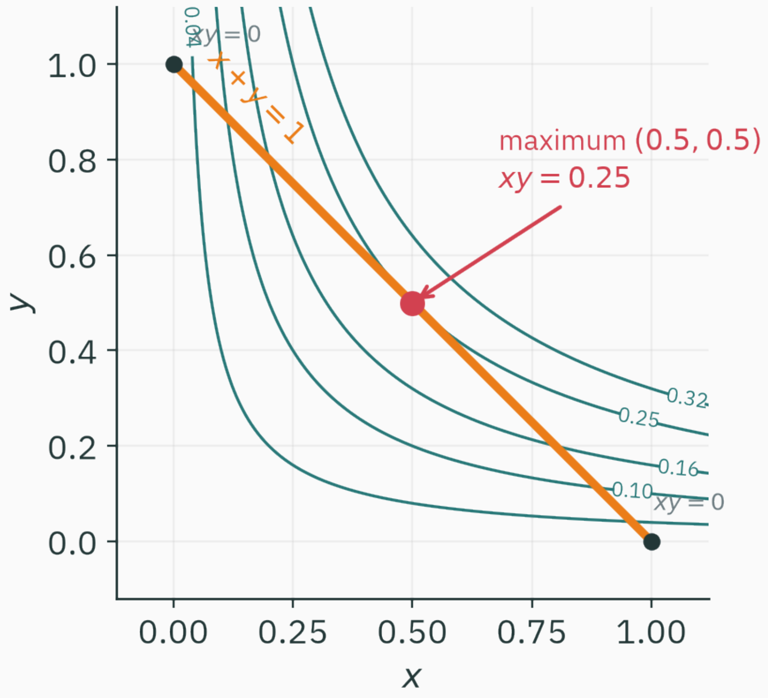
For $f(x, y) = xy$ at $(0.5, 0.5)$,

$$\nabla f = \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The gradient points uphill, but that direction leaves the line $x + y = 1$.

The feasible set is part of the problem

V VISUAL



The orange line contains every feasible point. The best contour that touches it does so at $(0.5, 0.5)$.

What should a general method return?

For the budget problem it should give:

1. the point $x^* = y^* = 0.5$,
2. the value $f^* = 0.25$,
3. a certificate that no small **feasible** move improves the objective,
4. and, if the budget changes, the rate at which the best value changes.

One extra scalar, the Lagrange multiplier, supplies both stationarity and sensitivity.

Learning outcomes

By the end of this lecture you will be able to:

1. derive ⭐ $\nabla f = \lambda \nabla h$ from feasible directions,
2. form a Lagrangian and solve its stationarity equations,
3. interpret λ^* as the local value of relaxing a constraint,
4. handle several equality constraints,
5. derive the maximum-entropy distribution on a finite set,
6. and distinguish a stationary candidate from a constrained optimum.

Geometry of an equality constraint

The constraint is a level set

Write the budget as

$$h(x, y) = x + y - 1 = 0.$$

The feasible set is the zero-level set of h.

Its gradient is

$$\nabla h = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

From L8, a gradient is normal to a level set. Therefore ∇h points across the feasible line, not along it.

Feasible directions are tangent directions

At a feasible point, let $\boldsymbol{v}$ be a small direction that stays on the constraint to first order.

$$h(\boldsymbol{x} + \varepsilon \boldsymbol{v}) \approx h(\boldsymbol{x}) + \varepsilon \nabla h(\boldsymbol{x})^\top \boldsymbol{v}.$$

Since $h(\boldsymbol{x}) = 0$, first-order feasibility requires

$$\nabla h(\boldsymbol{x})^\top \boldsymbol{v} = 0.$$

For $x + y = 1$, one tangent direction is $\boldsymbol{v} = (1, -1)$ because $(1, 1)^\top (1, -1) = 0$.

Stationarity only along the feasible set

At a constrained optimum, neither $\boldsymbol{v}$ nor $-\boldsymbol{v}$ may improve the objective to first order.

$f(\boldsymbol{x} + \varepsilon \boldsymbol{v}) \approx f(\boldsymbol{x}) + \varepsilon \nabla f(\boldsymbol{x})^\top \boldsymbol{v}.$

Hence every feasible tangent direction must satisfy

$\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{v} = 0.$

So $\nabla f(\boldsymbol{x}^*)$ is also normal to the feasible set.

⭐ Two normals to the same curve

D DERIVATION

For one regular constraint, the normal space has one direction: ∇h.

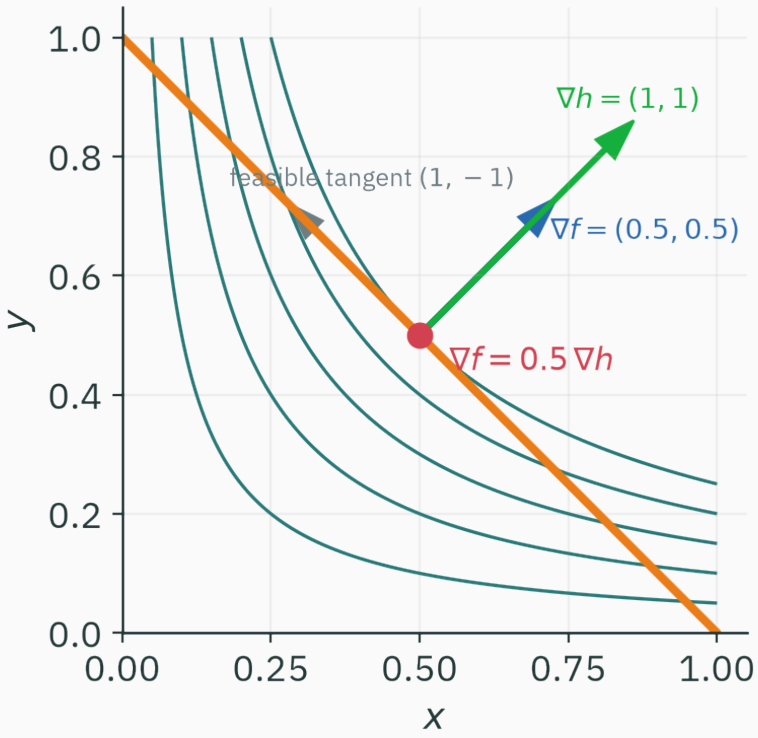
We just found that ∇f belongs to the same normal space.

Therefore some scalar λ satisfies

$$\boxed{\nabla f(\boldsymbol{x}^*) = \lambda^* \nabla h(\boldsymbol{x}^*).}$$

This is the Lagrange multiplier condition. "Regular" means $\nabla h(\boldsymbol{x}^*) \neq \mathbf{0}$.

The running example at the optimum



At $(0.5, 0.5)$,

$$\nabla f = (0.5, 0.5), \quad \nabla h = (1, 1), \quad \text{so} \quad \lambda^* = 0.5.$$

Check the tangent calculation directly

The feasible tangent direction is $\boldsymbol{v} = (1, -1)$.

At the candidate $(0.5, 0.5)$,

$\nabla f^\top \boldsymbol{v} = (0.5, 0.5)^\top (1, -1) = 0.$

At the feasible point $(0.2, 0.8)$,

$\nabla f^\top \boldsymbol{v} = (0.8, 0.2)^\top (1, -1) = 0.6.$

Moving a little toward increasing x and decreasing y improves the product there.

A one-variable check

The constraint gives $y = 1 - x$. Substitute it into the objective:

$$\varphi(x) = x(1-x) = x - x^2.$$

$$\varphi'(x) = 1 - 2x = 0 \quad \Rightarrow \quad x = \frac{1}{2}, \quad y = \frac{1}{2}.$$

$$\varphi''(x) = -2 < 0.$$

Therefore this candidate is a maximum.

> Substitution is an excellent check. Lagrange multipliers become useful when there are many variables or several coupled constraints.

Question: which statement is necessary?

Q QUESTION

At a regular equality-constrained optimum, which statement must hold?

A. $\nabla f = \mathbf{0}$

B. ∇f is parallel to ∇h

C. $f = h$

D. ∇f is tangent to $h = 0$

Answer

A ANSWER

Correct: B — ∇f is parallel to ∇h

Why: Both gradients are normal to the feasible level set at a regular constrained optimum.

The Lagrangian

Put the condition into one function

For a maximization problem

$$\max_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{subject to} \quad h(\boldsymbol{x}) = 0,$$

define

$$\mathcal{L}(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) - \lambda h(\boldsymbol{x}).$$

Stationarity with respect to both arguments gives

$$\nabla_{\boldsymbol{x}} \mathcal{L} = \nabla f - \lambda \nabla h = \mathbf{0},$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -h(\boldsymbol{x}) = 0.$$

The last equation puts the solution back on the constraint.

Sign conventions

Both definitions occur in books:

$$\mathcal{L} = f - \lambda h \quad \text{or} \quad \mathcal{L} = f + \lambda h.$$

Changing the sign changes the reported multiplier, not the point $\boldsymbol{x}^*$.

In this lecture, maximization examples use $\mathcal{L} = f - \lambda h$. State the convention before interpreting the sign of λ.

⭐ Step 1 — write the running Lagrangian

D DERIVATION

$$f(x, y) = xy, \quad h(x, y) = x + y - 1.$$

$$\mathcal{L}(x, y, \lambda) = xy - \lambda(x + y - 1).$$

There are now three unknowns and three stationarity equations.

⭐ Step 2 — differentiate

D DERIVATION

$$\frac{\partial \mathcal{L}}{\partial x} = y - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 1) = 0.$$

The first two equations say

$$x = y = \lambda.$$

⭐ Step 3 — enforce feasibility

D DERIVATION

Insert $x = y = \lambda$ into $x + y = 1$:

$$2\lambda = 1 \quad \Rightarrow \quad \lambda^* = \frac{1}{2}.$$

Therefore

$$\boxed{x^* = \tfrac{1}{2}, \quad y^* = \tfrac{1}{2}, \quad f(x^*, y^*) = \tfrac{1}{4}.}$$

The same calculation produced the optimizer and the multiplier.

The equations as a nonlinear system

The stationary point solves

$$F(x, y, \lambda) = \begin{pmatrix} y - \lambda \\ x - \lambda \\ x + y - 1 \end{pmatrix} = \mathbf{0}.$$

This viewpoint matters computationally: for harder problems we solve the coupled equations numerically.

```
from scipy.optimize import root

def F(z):
    x, y, lam = z
    return [y-lam, x-lam, x+y-1]

root(F, [0.3, 0.7, 0.0]).x
# array([0.5, 0.5, 0.5])
```

A minimum: closest point on a line

Find the point of smallest squared norm on $x + 2y = 4$:

$$\min_{x,y} \quad \frac{1}{2}(x^2 + y^2) \quad \text{subject to} \quad x + 2y - 4 = 0.$$

Use the minimization convention

$$\mathcal{L} = \frac{1}{2}(x^2 + y^2) + \lambda(x + 2y - 4).$$

Stationarity gives

$$x + \lambda = 0, \quad y + 2\lambda = 0, \quad x + 2y = 4.$$

Solve the closest-point example

$$x = -\lambda, \quad y = -2\lambda.$$

The constraint gives

$$-\lambda - 4\lambda = 4 \quad \Rightarrow \quad \lambda = -\frac{4}{5}.$$

$$\boxed{(x^*, y^*) = \left(\tfrac{4}{5}, \tfrac{8}{5}\right) = (0.8, 1.6).}$$

The point vector $(0.8, 1.6)$ is parallel to the line normal $(1, 2)$, as the geometry predicts.

Verify the answer numerically

```python
import numpy as np

x = np.array([0.8, 1.6])
print(round(x @ x / 2, 3))      # 1.6
print(np.array([1, 2]) @ x)     # 4.0

# nearby feasible direction v=(2,-1)
for t in [-0.2, 0.0, 0.2]:
    z = x + t*np.array([2, -1])
    print(t, round(z @ z / 2, 3))
# -0.2 1.7; 0.0 1.6; 0.2 1.7
```

Question: what does $\partial_\lambda \mathcal{L} = 0$ do?

Q QUESTION

In an equality-constrained Lagrangian, the multiplier equation...

A. minimizes the objective

B. sets the learning rate

C. enforces the original constraint

D. tests positive definiteness

Answer

A ANSWER

Correct: C — enforces the original constraint

Why: Differentiating with respect to the multiplier returns the constraint equation, up to sign.

What the multiplier measures

Let the budget change

Replace $x + y = 1$ by $x + y = b$:

$$V(b) = \max_{x,y} \quad xy \quad \text{subject to} \quad x + y = b.$$

The same stationarity equations give

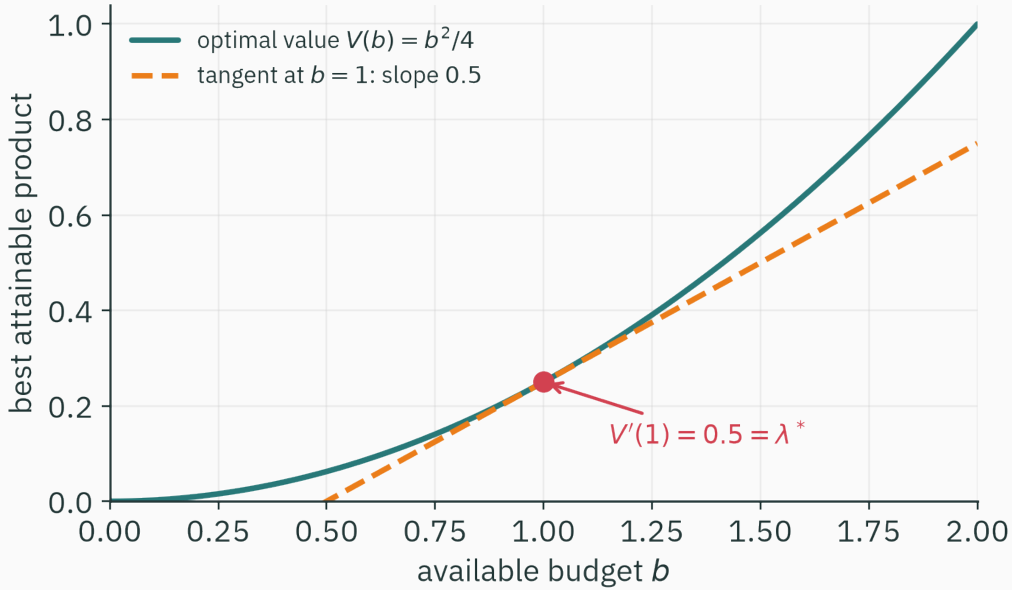
$$x^*(b) = y^*(b) = \frac{b}{2}.$$

Therefore the best attainable value is

$$V(b) = \frac{b^2}{4}.$$

The derivative of the optimal value

V VISUAL



At $b = 1$,

$$V'(1) = \frac{1}{2}.$$

The Lagrange calculation gave $\lambda^* = \frac{1}{2}$ as well.

Why the same number appears

Use

$$\mathcal{L}(x, y, \lambda; b) = xy - \lambda(x + y - b).$$

At the optimum, the derivatives through $x^*(b)$ and $y^*(b)$ vanish by stationarity. The remaining direct derivative is

$$\frac{\mathrm{d}V}{\mathrm{d}b} = \frac{\partial \mathcal{L}}{\partial b} = \lambda^*.$$

The multiplier is the local change in optimal value per unit relaxation of the constraint.

Read $\lambda = 0.5$ in units

Increase the budget from 1 to 1.1.

The first-order prediction is

$$V(1.1) \approx V(1) + \lambda^*(0.1) = 0.25 + 0.5(0.1) = 0.30.$$

The exact value is

$$V(1.1) = \frac{1.1^2}{4} = 0.3025.$$

The difference 0.0025 is a second-order effect.

Shadow prices in ML

constraint	multiplier	local reading
memory $= B$	λ_B	value of one more unit of memory
expected cost $= C$	λ_C	price of tightening the cost target
normalization $\sum p_i = 1$	λ	normalization offset in the optimum
fairness statistic $= 0$	λ_F	loss paid for local enforcement

The sign depends on how the constraint was written; the magnitude and units carry the interpretation.

Question: predict a relaxed budget

Q QUESTION

At $b = 1$, $V = 0.25$ and $\lambda^* = 0.5$. What is the first-order estimate of $V(1.02)$?

A. 0.2501

B. 0.26

C. 0.27

D. 0.75

Answer

A ANSWER

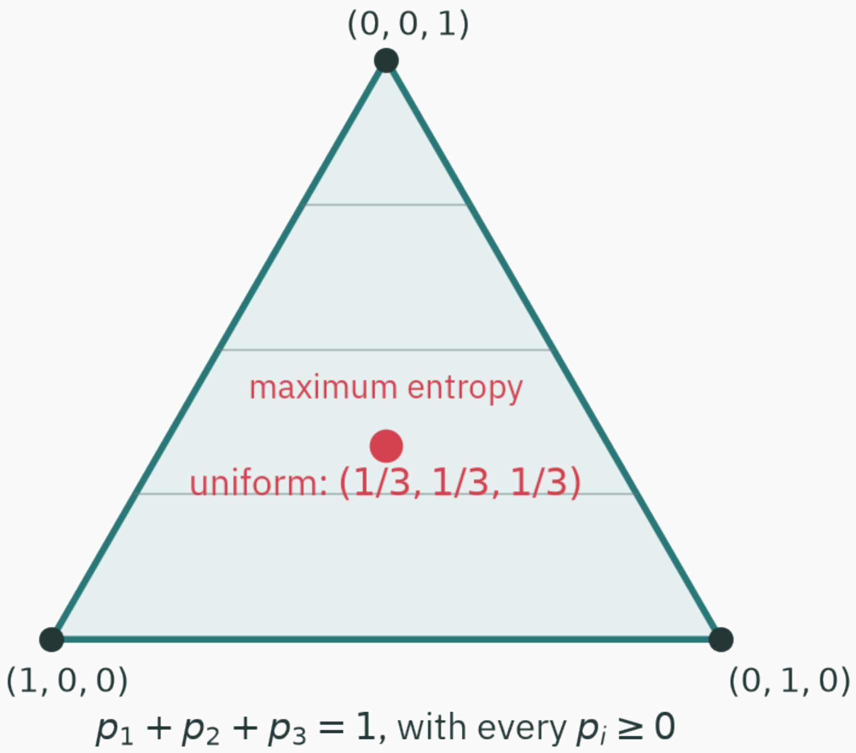
Correct: B — 0.26

Why: $V(1+\Delta b) \approx V(1) + \lambda^* \Delta b = 0.25 + 0.5(0.02)$.

A probability distribution with maximum entropy

Three probabilities live on a triangle

V VISUAL



Every distribution over three outcomes satisfies

$$p_1 + p_2 + p_3 = 1, \quad p_i \geq 0.$$

The equality removes one degree of freedom, so the feasible set is two-dimensional.

Entropy rewards spread

For a distribution $\boldsymbol{p}$ over K outcomes,

$$H(\boldsymbol{p}) = -\sum_{i=1}^{K} p_i \log p_i.$$

Compare three distributions for $K = 3$ (natural logarithm):

$\boldsymbol{p}$	$H(\boldsymbol{p})$
$(1, 0, 0)$	0
$(0.8, 0.1, 0.1)$	0.639
$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$	$\log 3 \approx 1.099$

Set up maximum entropy

$$\max_{\boldsymbol{p}} \quad -\sum_i p_i \log p_i \quad \text{subject to} \quad \sum_i p_i = 1.$$

For the moment, look for an interior candidate with $p_i > 0$. Define

$$\mathcal{L}(\boldsymbol{p}, \lambda) = -\sum_i p_i \log p_i - \lambda\left(\sum_i p_i - 1\right).$$

The inequality $p_i \geq 0$ will be handled systematically by KKT conditions in L20. L22 will verify that the interior candidate found here is the global maximum.

Differentiate one coordinate

For each i,

$$\frac{\partial \mathcal{L}}{\partial p_i} = -(\log p_i + 1) - \lambda = 0.$$

Hence

$$\log p_i = -1 - \lambda$$

and therefore

$$p_i = \exp(-1 - \lambda).$$

The right-hand side is the same for every i.

Enforce normalization

All K probabilities are equal. Write each one as c.

$$\sum_{i=1}^{K} p_i = Kc = 1 \quad \Rightarrow \quad c = \frac{1}{K}.$$

Maximum entropy on K outcomes: $p_i^* = \frac{1}{K}$, with $H(\boldsymbol{p}^*) = \log K$.

For $K = 3$, the solution is the centre of the triangle in the previous figure.

The multiplier comes from $\log p_i = -1 - \lambda$: here $\lambda^* = \log K - 1$ — the sensitivity of the maximal entropy to the total-probability budget, exactly as $V'(b) = \lambda^*$ before.

Check the entropy calculation in code

```
import numpy as np

def entropy(p):
    p = np.asarray(p)
    p = p[p > 0]
    return (p*np.log(1/p)).sum()

for p in ([1, 0, 0], [.8, .1, .1], [1/3]*3):
    print(round(entropy(p), 3))
# 0.0
# 0.639
# 1.099
```

Why this example matters

The probability simplex appears whenever a model produces categorical probabilities.

1. **Softmax** maps logits to the interior of this simplex.
2. **Cross-entropy** compares a predicted distribution with a target.
3. **Maximum entropy** chooses the least concentrated distribution consistent with known constraints.
4. Extra moment constraints lead to exponential-family distributions (⭐⭐⭐ slide at the end).

This calculation will return in the information-theory module.

Several equality constraints

One multiplier per constraint

For

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{subject to} \quad h_j(\boldsymbol{x}) = 0, \quad j = 1, ..., m,$$

define

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_{j=1}^{m} \lambda_j h_j(\boldsymbol{x}).$$

Stationarity becomes

$$\nabla f(\boldsymbol{x}^*) + \sum_j \lambda_j^* \nabla h_j(\boldsymbol{x}^*) = \mathbf{0},$$

$$h_j(\boldsymbol{x}^*) = 0 \quad \text{for every } j.$$

The gradient lies in a span

With one equality, ∇f is parallel to one constraint gradient.

With several equalities,

$$-\nabla f(\boldsymbol{x}^*) \in \operatorname{span}\{\nabla h_1(\boldsymbol{x}^*), \ldots, \nabla h_m(\boldsymbol{x}^*)\}.$$

The multipliers are the coordinates of $-\nabla f$ in this normal space.

> The feasible tangent space is orthogonal to every constraint gradient. At stationarity, the objective has zero component in that tangent space.

Linear equalities in matrix form

For $A\boldsymbol{x} = \boldsymbol{b}$,

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top (A\boldsymbol{x} - \boldsymbol{b}).$$

The stationarity equations are

$$\nabla f(\boldsymbol{x}) + A^\top \boldsymbol{\lambda} = \boldsymbol{0},$$

$$A\boldsymbol{x} = \boldsymbol{b}.$$

These equations are called the **KKT system** for an equality-constrained problem. L20 adds inequalities.

Projection onto linear constraints

Project a point $\boldsymbol{z}$ onto $A\boldsymbol{x} = \boldsymbol{b}$:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{z}\|^2 \quad \text{subject to} \quad A\boldsymbol{x} = \boldsymbol{b}.$$

Its Lagrangian is

$$\mathcal{L} = \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{z}\|^2 + \boldsymbol{\lambda}^\top (A\boldsymbol{x} - \boldsymbol{b}).$$

⭐ Solve the projection equations

D DERIVATION

Stationarity in $\boldsymbol{x}$ gives

$$\boldsymbol{x} - \boldsymbol{z} + A^\top \boldsymbol{\lambda} = \boldsymbol{0},$$

so

$$\boldsymbol{x} = \boldsymbol{z} - A^\top \boldsymbol{\lambda}.$$

Insert this into $A\boldsymbol{x} = \boldsymbol{b}$:

$$A\boldsymbol{z} - AA^\top \boldsymbol{\lambda} = \boldsymbol{b}.$$

⭐ The projection formula

D DERIVATION

If $AA^\top$ is invertible,

$$\boldsymbol{\lambda} = (AA^\top)^{-1}(A\boldsymbol{z} - \boldsymbol{b}).$$

Therefore

$$\boxed{\boldsymbol{x}^* = \boldsymbol{z} - A^\top(AA^\top)^{-1}(A\boldsymbol{z} - \boldsymbol{b}).}$$

The correction lies in the row space of A—normal to the feasible affine set.

Numbers: project $(2, 0)$ onto $x + y = 1$

$$\boldsymbol{z} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad A = (1 \ \ 1), \quad b = 1.$$

$$A\boldsymbol{z} - b = 1, \quad AA^\top = 2.$$

$$\boldsymbol{x}^* = \begin{pmatrix} 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \left(\frac{1}{2} \right) (1) = \begin{pmatrix} 1.5 \\ -0.5 \end{pmatrix}.$$

Check: $1.5 + (-0.5) = 1$, and the correction $(-0.5, -0.5)$ is normal to the line.

The same projection in NumPy

```python
import numpy as np

z = np.array([2., 0.])
A = np.array([[1., 1.]])
b = np.array([1.])

lam = np.linalg.solve(A @ A.T, A @ z - b)
x = z - A.T @ lam
print(x)        # [ 1.5 -0.5]
print(A @ x)    # [1.]
```

As in L4, solve the linear system; do not form a matrix inverse in code.

Candidates, not guarantees

Stationarity is necessary, not sufficient

Consider

$$\max_{x,y} \quad x^2 \quad \text{subject to} \quad x^2 + y^2 = 1.$$

The Lagrange equations return four stationary points:

$$(\pm 1, 0) \quad \text{and} \quad (0, \pm 1).$$

But their objective values are different:

$$f(\pm 1, 0) = 1 \quad (\text{maxima}), \quad f(0, \pm 1) = 0 \quad (\text{minima}).$$

We still have to compare candidates or inspect second-order behaviour along feasible directions.

Regularity can fail

The geometric derivation assumed $\nabla h(\boldsymbol{x}^*) \neq \mathbf{0}$.

For the constraint $h(x) = x^2 = 0$, the only feasible point is $x = 0$, but

$h'(0) = 0.$

The feasible set is perfectly clear; the gradient description is not regular there.

> Constraint qualifications state when multiplier conditions are valid. L20 uses one of them when inequalities become active.

Equality constraints do not describe boundaries

Many ML problems also contain inequalities:

$$p_i \geq 0, \quad \|\boldsymbol{w}\| \leq R, \quad \text{loss}_g \leq \varepsilon, \quad x_i \geq 0.$$

An inequality may be:

1. inactive, with room to spare, or
2. active, behaving like an equality at the solution.

KKT conditions decide which case applies. That is L20.

A compact procedure

For a smooth equality-constrained problem:

1. Write every equality as $h_j(\boldsymbol{x}) = 0$.
2. Form $\mathcal{L} = f + \sum_j \lambda_j h_j$; record the sign convention.
3. Solve $\nabla_{\boldsymbol{x}} \mathcal{L} = \mathbf{0}$ and $h_j(\boldsymbol{x}) = 0$ together.
4. Check feasibility and compare all candidates.
5. Use curvature, convexity, or direct comparison to classify them.
6. Interpret λ_j^* only after fixing the units and sign.

Summary

At a regular equality-constrained optimum, the objective gradient lies in the span of the constraint gradients.

1. The Lagrangian turns that geometric condition and the constraints into one system of equations.
2. A multiplier measures local sensitivity of the optimal value to its constraint.
3. One multiplier is introduced for each equality.
4. The probability simplex gives a central ML example: entropy is maximized by the uniform distribution.
5. Lagrange equations produce stationary candidates; they do not by themselves prove optimality.

Practice

1. Minimize $x^2 + y^2$ subject to $x + y = 6$. Give (x^*, y^*) and λ^* under $\mathcal{L} = f + \lambda(x + y - 6)$.
2. Maximize xyz subject to $x + y + z = 12$ and $x, y, z > 0$.
3. Project $\boldsymbol{z} = (3, 1)$ onto $2x - y = 0$ using the matrix projection formula.
4. For maximum entropy with $K = 5$, find $\boldsymbol{p}^*$ and $H(\boldsymbol{p}^*)$.

Read before L20 — MML §7.2 (equality constraints); Boyd & Vandenberghe §5.1, gently — the pictures and boxed statements first.

Practice answers

A ANSWER

1. $(x^*, y^*) = (3, 3)$ and $\lambda^* = -6$.
2. $(x^*, y^*, z^*) = (4, 4, 4)$ and the maximum product is 64.
3. $A = (2, -1)$, so $A\boldsymbol{z} = 5$, $AA^\top = 5$, and $\boldsymbol{x}^* = \boldsymbol{z} - A^\top\left(AA^\top\right)^{-1} A\boldsymbol{z} = (1, 2)$.
4. $p_i^* = \frac{1}{5}$ and $H = \log 5$ nats.

⭐⭐⭐ Add one expected-value constraint

★ OPTIONAL

Suppose outcomes have values a_i, and we also require $\sum_i p_i a_i = \mu$.

$$\mathcal{L} = -\sum_i p_i \log p_i - \lambda\left(\sum_i p_i - 1\right) - \beta\left(\sum_i p_i a_i - \mu\right).$$

Stationarity gives

$$p_i = \exp(-1 - \lambda - \beta a_i) \propto \exp(-\beta a_i).$$

Normalization determines the missing constant. This is the basic exponential-family form.

Next: inequalities

The probability simplex already contained $p_i \geq 0$, which we temporarily set aside.

L20 keeps the running example — maximize xy with the budget as an **inequality**, $x + y \leq 1$ — and asks when a constraint actually binds.

Next lecture:

$$\text{primal feasibility} + \text{dual feasibility} + \text{stationarity} + \text{complementary slackness}.$$

Together these are the Karush–Kuhn–Tucker conditions.

At the optimum, the objective and the constraint pull in the same direction.

Next: **Duality & KKT** — inequality constraints, and a certificate that no better feasible point exists.